Data Structures and Algorithms

Project Proposal

Bus Reservation System

Section: A

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Submitted To:

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Problem:

The current public transportation system faces challenges in efficiently managing bookings, minimizing overbooking, and ensuring a smooth user experience for passengers. Traditional systems often lack features like real-time seat availability, route optimization, and a user-friendly interface, which are crucial for enhancing user satisfaction and operational efficiency.

Passengers also face difficulties in reserving seats in advance, canceling bookings, and viewing available routes dynamically. Furthermore, the system administrators struggle to effectively manage bus schedules and customer data, which leads to inefficiencies and delays.

Significance and Relevance:

An efficient Bus Reservation System is critical for ensuring smooth operations in public transportation. It will use advanced data structures and algorithms to solve real-world problems like route optimization, seat allocation, and real-time booking management. This project will demonstrate the practical application of DSA concepts like graphs, queues, heaps, and trees to enhance system performance.

The implementation will also ensure the security of user data and provide a seamless experience for passengers and administrators alike.

Features:

- **Real-Time Booking**
- Enables passengers to view and book seats in real-time with instant updates
- **Route Optimization**
  Utilizes graph algorithms to find the shortest and most efficient routes.
- **Secure Payment System**
  Integrates secure methods to handle transactions, ensuring user trust.
- **Dynamic Seat Allocation**
  Implements data structures like priority queues to efficiently manage seat availability.
- **Passenger Dashboard**
  Allows passengers to track bookings, view travel history, and manage cancellations.
- **Admin Panel**
  Enables administrators to manage bus schedules, passenger data, and revenue reports.
- **Notifications and Alerts**
  Sends alerts for delays, cancellations, or other critical updates.
- **Categorized Routes**
  Uses tree-based structures to manage route hierarchies and their schedules.

Members and their contributions:

- **Shoaib Khalid**

Passenger functionality (real-time booking, dashboard, and history).

- **Tahir Nawaz**

Admin functionality (schedule management, revenue reporting).

- **Mohsin Ali**

Algorithms for route optimization and seat allocation.